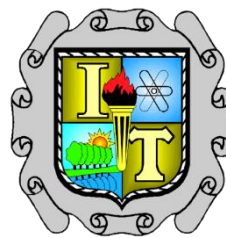




**TECNOLÓGICO
NACIONAL DE MÉXICO**



Tecnológico Nacional de México

Instituto Tecnológico de Saltillo

Ingeniería en Sistemas Computacionales

Cómputo en la Nube

Unidad 4

Práctica 4.10 – Análisis de tráfico de red con Wireshark

Alumna:

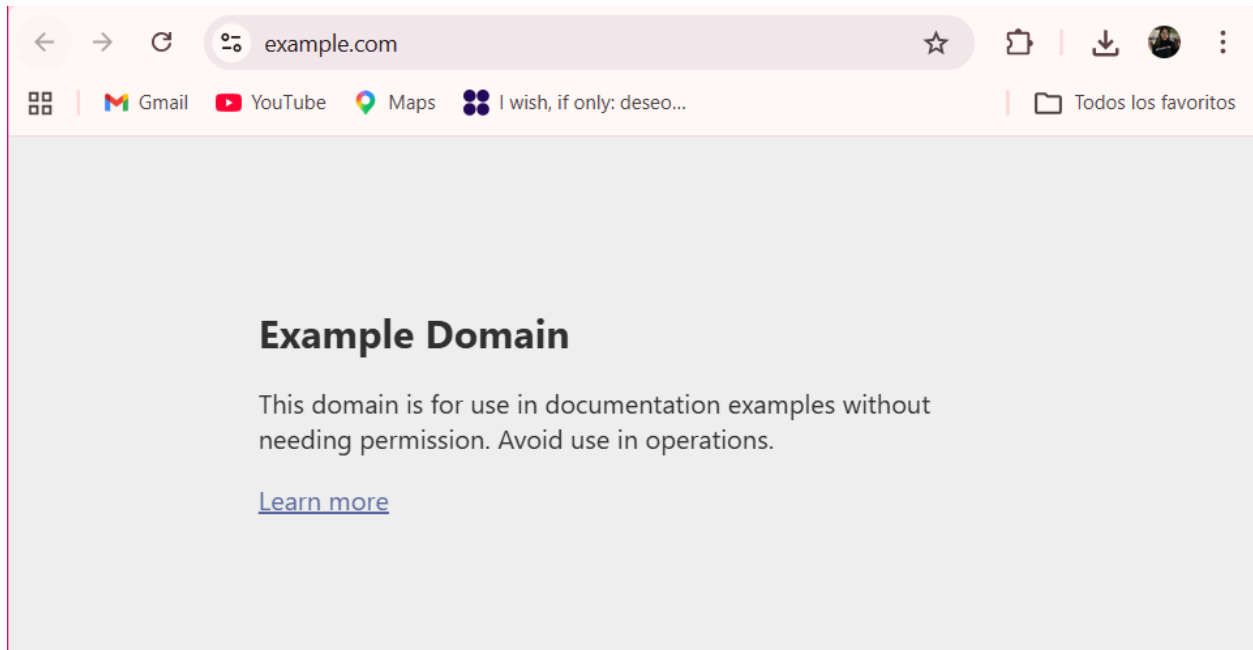
Silvia Guadalupe Ramos Mendoza 22050726

Docente: Ing. Miguel Salazar del Bosque

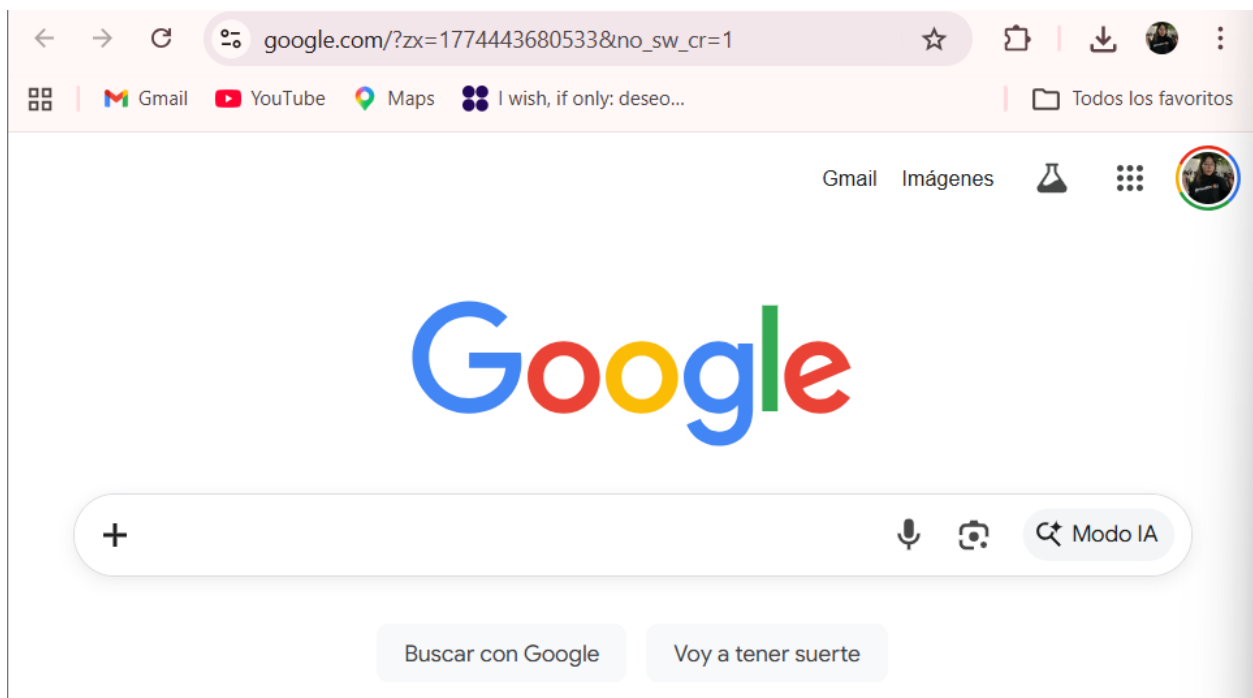
Saltillo, Coahuila; a 25 de marzo de 2026

Abrir Links:

<http://example.com>



<https://google.com>



Filtrar tráfico HTTP

The screenshot shows the Wi-Fi network analyzer interface with the 'http' filter applied. The packet list displays two HTTP GET requests to 'cert.ssl.com'. The selected packet (No. 1594) is expanded, showing the Hypertext Transfer Protocol details, including the Host 'cert.ssl.com' and the full request URI 'http://cert.ssl.com/SSL.com-TLS-T-ECC-R2.cer'. The packet details pane on the right shows the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
1564	31.744159900	10.0.42.195	16.15.212.4	HTTP	352	GET /SSL.com-TLS-T-ECC-R2.cer HTTP/1.1
1594	31.968317400	16.15.212.4	10.0.42.195	HTTP	878	HTTP/1.1 200 OK (application/pkix-cert)

Character encoding (ASCII, EBCDIC...) (frame.encoding) | Packets: 5296 - Displayed: 2 (0.0%) - Dropped: 0 (0.0%) | Profile: Default

Filtrar tráfico DNS

The screenshot shows the Wi-Fi network analyzer interface with the 'dns' filter applied. The packet list displays several DNS queries and responses. The selected packet (No. 350) is expanded, showing the Domain Name System (query) details, including the Transaction ID '0xde93' and the query for 'storeedgefd.dsx.mp.microsoft.com'. The packet details pane on the right shows the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
350	12.265955000	10.0.42.195	8.8.8.8	DNS	92	Standard query 0xde93 A storeedgefd.dsx.mp.microsoft.com
358	12.340998100	8.8.8.8	10.0.42.195	DNS	320	Standard query response 0xde93 A storeedgefd.dsx.mp.microsoft.com CNAME storeedgefd.xbetservices.akadns.net CNAME storeedgefd.dsx.mp.microsoft.com
454	14.824342100	10.0.42.195	8.8.8.8	DNS	70	Standard query 0x1761 HTTPS dns.google
455	14.824559700	10.0.42.195	8.8.8.8	DNS	70	Standard query 0xbab9 A dns.google
483	14.857298800	8.8.8.8	10.0.42.195	DNS	150	Standard query response 0x1761 HTTPS dns.google SOA ns1.zdns.google
488	14.870703000	8.8.8.8	10.0.42.195	DNS	102	Standard query response 0xbab9 A dns.google A 8.8.4.4 A 8.8.8.8
642	15.734207400	10.0.42.195	8.8.8.8	DNS	89	Standard query 0x1a45 A v10.events.data.microsoft.com
645	15.778810500	8.8.8.8	10.0.42.195	DNS	233	Standard query response 0x1a45 A v10.events.data.microsoft.com CNAME win-global-asimov-leafs-events-data.trafficmanager.net CNAME win-global-asimov-leafs-events-data.trafficmanager.net
683	15.842814600	10.0.42.195	8.8.8.8	DNS	75	Standard query 0x803f A upad.mshome.net
688	15.913183400	8.8.8.8	10.0.42.195	DNS	161	Standard query response 0x803f No such name A upad.mshome.net SOA ns1-35.azure-dns.com

Domain Name System: Protocol | Packets: 5296 - Displayed: 38 (0.7%) - Dropped: 0 (0.0%) | Profile: Default

Filtrar tráfico TCP

Wireshark capture of TCP traffic. The packet list shows a GET request to a Fortinet SSL certificate. The packet details pane shows the full request URI: <http://cert.ssl.com/SSL.com-TLS-T-ECC-R2.cer>. The packet bytes pane shows the raw data.

Frame 1564: Packet, 352 bytes on wire (2816 bits), 352 bytes captured (2816 bits) on interface \Device\NPF_{6CA5A65A-DA18-4047-A198-C6E8431B26} Ethernet II, Src: Intel_98:0e:68 (b0:3c:dc:98:0e:68), Dst: Fortinet_ab:34:07 (e8:1c:ba:ab:34:07)

Destination: Fortinet_ab:34:07 (e8:1c:ba:ab:34:07)

Source: Intel_98:0e:68 (b0:3c:dc:98:0e:68)

Type: IPv4 (0x0800)

[Stream index: 23]

Internet Protocol Version 4, Src: 10.0.42.195, Dst: 16.15.212.4

Transmission Control Protocol, Src Port: 63597, Dst Port: 80, Seq: 1, Ack: 1, Len: 298

Hypertext Transfer Protocol

GET /SSL.com-TLS-T-ECC-R2.cer HTTP/1.1\r\n

Host: cert.ssl.com\r\n

Connection: keep-alive\r\n

User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36\r\n

Accept-Encoding: gzip, deflate\r\n

Accept-Language: es-US,es-419;q=0.9,es;q=0.8,en;q=0.7\r\n

\r\n

[Response in frame: 1594]

[Full request URI: <http://cert.ssl.com/SSL.com-TLS-T-ECC-R2.cer>]

1er. Sitio Visitado: <https://www.taylorswift.com/>

Wireshark capture of HTTP traffic. The packet list shows a GET request to a DigiCert SSL certificate. The packet details pane shows the full request URI: <http://cr13.digicert.com/DigiCertTrustedG4CodeSigningRSA4096SHA3842021CA1.crl>. The packet bytes pane shows the raw data.

Frame 1211: Packet, 338 bytes on wire (2704 bits), 338 bytes captured (2704 bits) on interface \Device\NPF_{6CA5A65A-DA18-4047-A198-C6E8431B26} Ethernet II, Src: Intel_98:0e:68 (b0:3c:dc:98:0e:68), Dst: Fortinet_ab:34:07 (e8:1c:ba:ab:34:07)

Destination: Fortinet_ab:34:07 (e8:1c:ba:ab:34:07)

Source: Intel_98:0e:68 (b0:3c:dc:98:0e:68)

Type: IPv4 (0x0800)

[Stream index: 2]

Internet Protocol Version 4, Src: 10.0.42.195, Dst: 23.11.33.159

Transmission Control Protocol, Src Port: 58037, Dst Port: 80, Seq: 1, Ack: 1, Len: 284

Hypertext Transfer Protocol

GET /DigiCertTrustedG4CodeSigningRSA4096SHA3842021CA1.crl HTTP/1.1\r\n

Cache-Control: max-age = 1378\r\n

Connection: Keep-Alive\r\n

Accept: */*\r\n

If-Modified-Since: Tue, 24 Mar 2026 20:05:16 GMT\r\n

If-None-Match: "69c2ee7c-27d35"\r\n

User-Agent: Microsoft-CryptoAPI/10.0\r\n

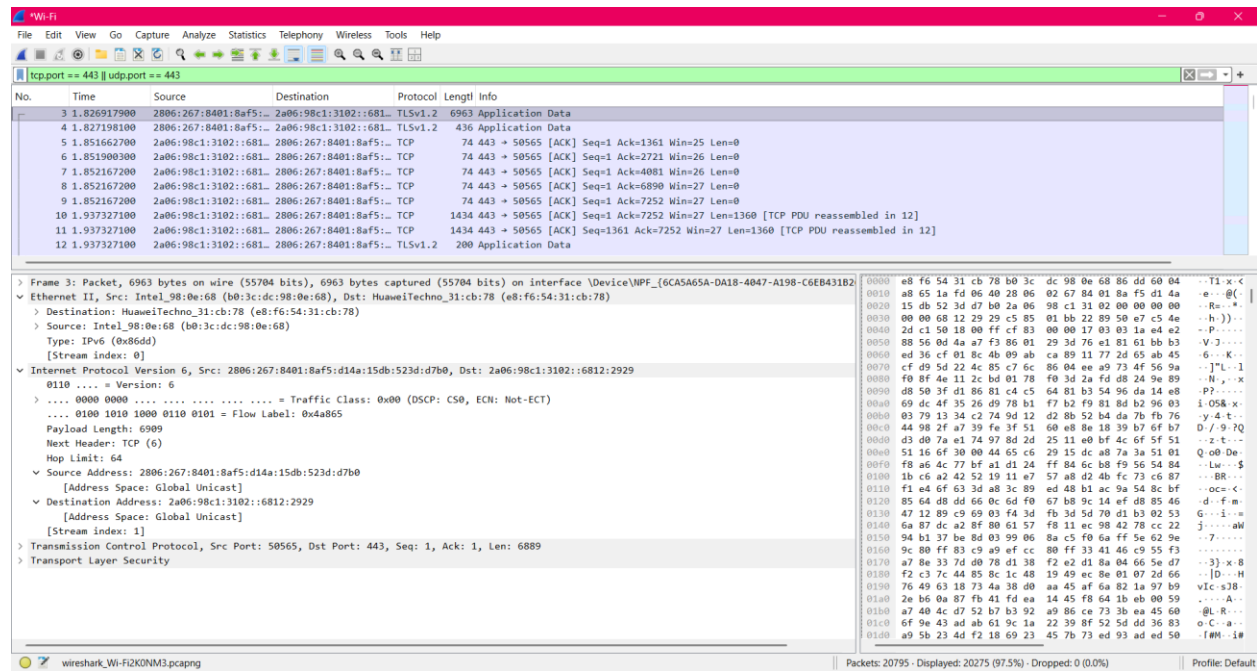
Host: cr13.digicert.com\r\n

\r\n

[Response in frame: 1343]

[Full request URI: <http://cr13.digicert.com/DigiCertTrustedG4CodeSigningRSA4096SHA3842021CA1.crl>]

2do. Sitio Visitado: <https://udiscover.mx/>



Wireshark capture showing traffic to udiscover.mx. The capture is on interface \Device\NPF_{6C5A65A-DA18-4047-A198-C6EB431B2603}. The selected packet is a TCP Reset (RST) from 2086:267:8401:8af5::2a06:98c1:3102::681 to 2086:267:8401:8af5::2a06:98c1:3102::681, port 443. The packet details show the Ethernet II header, Internet Protocol Version 6, and Transmission Control Protocol (RST). The packet bytes show the raw data.

Frame 3: Packet, 6963 bytes on wire (55704 bits), 6963 bytes captured (55704 bits) on interface \Device\NPF_{6C5A65A-DA18-4047-A198-C6EB431B2603}

Ethernet II, Src: Intel_98:0e:68 (b0:3c:dc:98:0e:68), Dst: HuaweiTechno_31:cb:78 (e8:f6:54:31:cb:78)

Destination: HuaweiTechno_31:cb:78 (e8:f6:54:31:cb:78)

Source: Intel_98:0e:68 (b0:3c:dc:98:0e:68)

Type: IPv6 (0x86dd)

[Stream index: 0]

Internet Protocol Version 6, Src: 2086:267:8401:8af5:d14a:15db:523d:d7b0, Dst: 2a06:98c1:3102::681:2929

..... = Version: 6

..... = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)

..... 0100 1010 1000 0101 = Flow Label: 0xa8a65

Payload Length: 6909

Next Header: TCP (6)

Hop Limit: 64

Source Address: 2086:267:8401:8af5:d14a:15db:523d:d7b0

[Address Space: Global Unicast]

Destination Address: 2a06:98c1:3102::681:2929

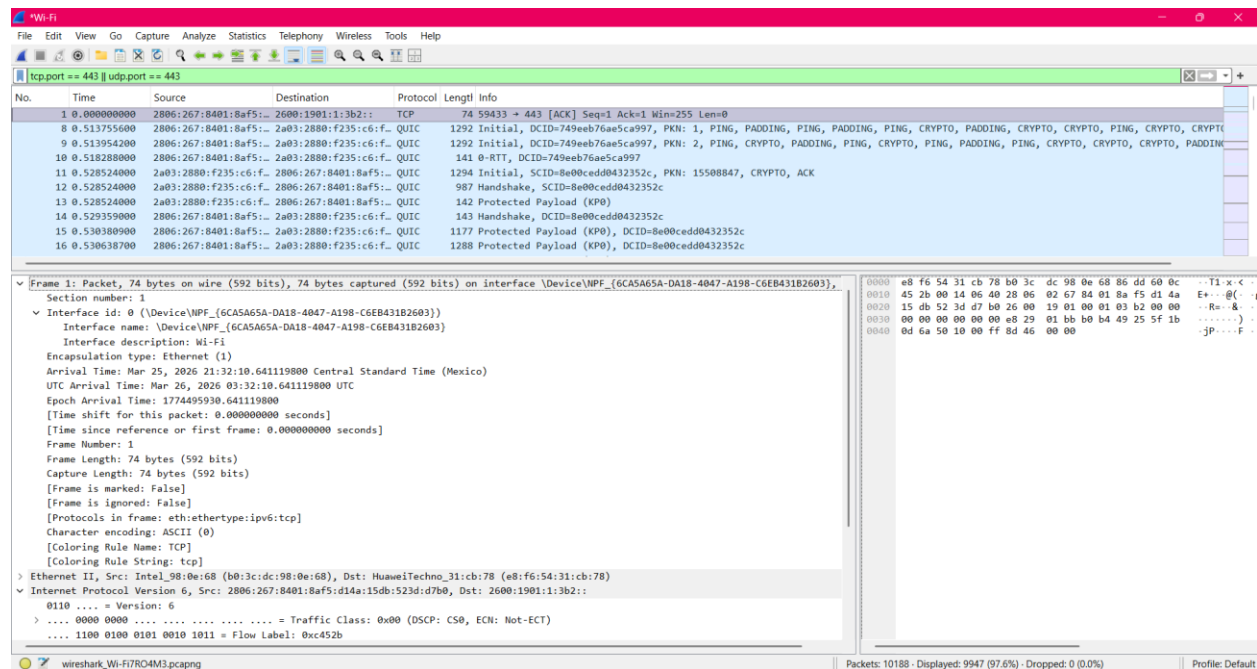
[Address Space: Global Unicast]

[Stream index: 1]

Transmission Control Protocol, Src Port: 50565, Dst Port: 443, Seq: 1, Ack: 1, Len: 6889

Transport Layer Security

3er. Sitio Visitado: <https://teams.cloud.microsoft/>



Wireshark capture showing traffic to teams.cloud.microsoft. The capture is on interface \Device\NPF_{6C5A65A-DA18-4047-A198-C6EB431B2603}. The selected packet is a TCP Reset (RST) from 2086:267:8401:8af5::2a06:98c1:3102::681 to 2086:267:8401:8af5::2a06:98c1:3102::681, port 443. The packet details show the Ethernet II header, Internet Protocol Version 6, and Transmission Control Protocol (RST). The packet bytes show the raw data.

Frame 1: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{6C5A65A-DA18-4047-A198-C6EB431B2603}, Section number: 1

Interface id: 0 (\Device\NPF_{6C5A65A-DA18-4047-A198-C6EB431B2603})

Interface name: \Device\NPF_{6C5A65A-DA18-4047-A198-C6EB431B2603}

Interface description: Wi-Fi

Encapsulation type: Ethernet (1)

Arrival Time: Mar 25, 2026 21:32:10.641119800 Central Standard Time (Mexico)

UTC Arrival Time: Mar 26, 2026 03:32:10.641119800 UTC

Epoch Arrival Time: 1774495930.641119800

[Time shift for this packet: 0.000000000 seconds]

[Time since reference or first frame: 0.000000000 seconds]

Frame Number: 1

Frame Length: 74 bytes (592 bits)

Capture Length: 74 bytes (592 bits)

[Frame is marked: False]

[Protocols in frame: ethertype:ipv6:tcp]

Character encoding: ASCII (0)

[Coloring Rule Name: TCP]

[Coloring Rule String: tcp]

Ethernet II, Src: Intel_98:0e:68 (b0:3c:dc:98:0e:68), Dst: HuaweiTechno_31:cb:78 (e8:f6:54:31:cb:78)

Internet Protocol Version 6, Src: 2086:267:8401:8af5:d14a:15db:523d:d7b0, Dst: 2086:267:8401:8af5::2a06:98c1:3102::681

..... = Version: 6

..... = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)

..... 1100 0100 0101 0010 1011 = Flow Label: 0xc452b

IP de otra laptop: 192.168.100.6

The image shows a Wireshark capture of network traffic. The top pane displays a list of captured packets, with three SSDP M-SEARCH requests from 192.168.100.6 to 239.255.255.250. The bottom pane shows the details of the selected packet (No. 632), including Ethernet II, Internet Protocol Version 4, User Datagram Protocol, and Simple Service Discovery Protocol layers. The packet data is displayed in hexadecimal and ASCII format on the right.

No.	Time	Source	Destination	Protocol	Length	Info
632	38.357413000	192.168.100.6	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
668	41.326671500	192.168.100.6	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
672	44.399309000	192.168.100.6	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1

Section number: 1
Interface id: 0 (\\Device\\NPF_{6C5A65A-DA18-4047-A198-C6EB431B2603})
Interface name: \\Device\\NPF_{6C5A65A-DA18-4047-A198-C6EB431B2603}
Interface description: Wi-Fi
Encapsulation type: Ethernet (1)
Arrival Time: Mar 25, 2026 21:53:22.282718900 Central Standard Time (Mexico)
UTC Arrival Time: Mar 26, 2026 03:53:22.282718900 UTC
Epoch Arrival Time: 1774497202.282718900
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 634.048600 milliseconds]
[Time since reference or first frame: 38.357413000 seconds]
Frame Number: 632
Frame Length: 179 bytes (1432 bits)
Capture Length: 179 bytes (1432 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:udp:ssdp]
Character encoding: ASCII (0)
[Coloring Rule Name: UDP]
[Coloring Rule String: udp]
> Ethernet II, Src: Intel_87:13:9b (e0:d4:64:e7:13:9b), Dst: IPv4mcast_7f:ff:fa (01:00:5e:7f:ff:fa)
> Internet Protocol Version 4, Src: 192.168.100.6, Dst: 239.255.255.250
> User Datagram Protocol, Src Port: 63536, Dst Port: 1900
> Simple Service Discovery Protocol

TCP Stream

The image shows the Wireshark TCP Stream window, displaying the data of a selected packet (No. 969). The window shows the raw data of the packet, including the HTTP request and response. The request is a GET /favicon.ico HTTP/1.1 from 192.168.100.6 to 192.168.100.6. The response is a 200 OK from 192.168.100.6 to 192.168.100.6. The window also shows the packet details, including the Ethernet II, Internet Protocol Version 4, and User Datagram Protocol layers.

GET /favicon.ico HTTP/1.1
Host: example.com
Connection: keep-alive
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Accept-Encoding: gzip, deflate
Accept-Language: es-US,es-419;q=0.9,es;q=0.8,en;q=0.7

HTTP/1.1 200 OK
Date: Thu, 26 Mar 2026 03:56:10 GMT
Content-Type: text/html
Transfer-Encoding: chunked
Connection: keep-alive
Content-Encoding: gzip
Last-Modified: Tue, 24 Mar 2026 22:06:31 GMT
Allow: GET, HEAD
Age: 1443
cf-cache-status: HIT
Vary: Accept-Encoding
Server: cloudflare
CF-RAY: 9e2339548cbd6786-DFW

GET /favicon.ico HTTP/1.1
Host: example.com
Connection: keep-alive
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36
Accept: image/avif,image/webp,image/apng,image/svg+xml,image/*,*/*;q=0.8
Referer: http://example.com/

¿Qué información se ve en HTTP?

Aquí todo es visible: La URL completa, métodos de la petición, cabeceras HTTP, cuerpo del mensaje, credenciales. Un atacante en la red puede leer contraseñas, cookies, datos privados.

¿Qué NO se ve en HTTPS?

No se puede ver la URL completa (solo el dominio a veces), tampoco puede verse el contenido de la página o los formularios enviados. De igual forma, no se pueden ver cookies ni headers internos, así como datos transmitidos.

En HTTP, toda la información viaja sin cifrado, lo que permite visualizar datos como la URL completa, encabezados y contenido en texto claro. En cambio, HTTPS utiliza cifrado, por lo que no es posible ver el contenido de la comunicación ni los datos sensibles, únicamente información básica como la IP y el dominio. Esto demuestra que HTTPS es mucho más seguro para la transmisión de información.

¿Qué información es visible?

Durante el análisis del tráfico de red con Wireshark, fue posible observar diversa información en texto claro, especialmente en el protocolo HTTP. Entre los datos visibles se identificaron:

- El método de la petición (GET), que indica el tipo de solicitud realizada.
- La URL completa del recurso solicitado, incluyendo el dominio y la ruta.
- El host del servidor al que se realiza la conexión.
- Información del navegador del usuario (User-Agent).
- Idioma del sistema (Accept-Language).
- Tipo de conexión y otros encabezados (headers).
- Direcciones IP de origen y destino, lo que permite identificar el equipo local y el servidor externo.
- Consultas DNS, donde se observa la relación entre nombres de dominio y direcciones IP.

¿Qué riesgo existe?

El principal riesgo identificado es la exposición de información sensible debido a la falta de cifrado en el protocolo HTTP. Al ser visible en texto claro, un atacante conectado a la misma red podría interceptar los datos transmitidos.

Entre los riesgos más importantes se encuentran:

- Intercepción de información personal o confidencial.
- Visualización de sitios web visitados por el usuario.
- Posibilidad de robo de credenciales si se envían sin cifrado.
- Análisis del comportamiento del usuario en la red.
- Ataques como “Man in the Middle”, donde un tercero puede espiar o modificar la comunicación.

Esto resalta la importancia del uso de protocolos seguros como HTTPS, los cuales cifran la información y reducen significativamente estos riesgos.

Conclusiones

En esta práctica pude observar cómo funciona el tráfico de red en tiempo real utilizando Wireshark, identificando información visible en protocolos como HTTP y DNS. Me di cuenta de que, cuando no hay cifrado, muchos datos pueden verse fácilmente, lo que representa un riesgo para la seguridad. Esto me ayudó a entender la importancia de utilizar protocolos seguros como HTTPS para proteger la información en la red.